COMSATS University Islamabad Lahore Campus

Mid-Term Examination – Fall 2023

Course Title:	Database Systems			Course Code:	CSC270	Credit	3+1
Course	Abdul Qayyum			Programme	BSCS		
Semester:	4th	Batch: SP22	Section:	B, C	Date:	01-11-23	
Time Allowed:	90 Minutes			Maximum Marks:		50	
Student`s				Reg. No.			

Important Instructions / Guidelines:

Question 1: [CLO-1] [5]

In which case we use cartesian product to retrieve data. Explain with an example.

Question 2: [CLO-2] [8]

Why we use Left outer and Right outer join to retrieve data from tables. Explain with an example.

Question 3: [CLO-2] [6]

What is the difference between base relation and database view?

Question 4: [CLO-2] [6]

Database System Development Lifecycle has many stages: Explain the following stages.

a. System definition
b. Prototyping
c. Data conversion and loading

Question 5: [CLO-3] [25]

Consider the following case to draw an ERD:

A General Hospital consists of a number of specialized wards (such as Maternity, Surgery, Oncology, etc). Each ward hosts a number of patients, who were admitted on the recommendation of their own family doctor and confirmed by a consultant employed by the Hospital. On admission, the personal details of every patient are recorded. A separate register is to be held to store the information of the tests undertaken and the results of a prescribed treatment. A number of tests may be conducted for each patient. Each patient is assigned to one leading consultant but may be examined by another doctor, if required. Doctors are specialists in some branch of medicine and may be leading consultants for a number of patients, not necessarily from the same ward.

Construct/Draw an ER diagram to represent the data requirements as following:

- Identify the main entity types: [07]
- Identify the main relationship types between the entity types. [05]
- Identify attributes and associate them with entity or relationship types: [05]
- Determine candidate and primary key attributes for each entity type. [03]
- Identify the occurrence (1 - 1, 1 - *, * - *) between entities. [05]
- State any assumptions necessary to support your design.

Moderated